Where does Qwen3-32B *hold* a paper's authors?

A layer-by-layer readout of author names for 9,108 papers

Stages 10–11 · jlens

Qwen3-32B · 2026-10-27

The question

target (4,621 papers)
can't write the list (R0/R1)
but picks the true list 4/4 times

control (2,220 papers)
can't write the list (R0/R1)
and can't pick the true list

If the model already **knows** target authors in its middle layers and only fails to say them, a lens on those layers should read the right name **better for target than control**.

If the knowledge is only assembled at the end, the gap should appear only near the output.

So: at which layer does target pull ahead of control?

The data

9,108 papers, 8 fields

33,811 author names

149,999 name tokens scored

13 citation bins (0 to 63,901)

Source: the **Dreaming Up Authors** evaluation set. 9,999,863 random OAG v2 records + Semantic Scholar metadata (Sep 2025) -> quality filters -> 1.92 M papers -> up to 100 per field x citation bin

Groups (group lead's labelling run, same model)

target 4,621 · control 2,220

guard (writes the authors) 214 · excluded 2,053

Medicine, biology, CS, psychology, chemistry, maths, physics, materials science: ~1,100 papers each

Name formats

73% start with an initial ("S. Bree")

26% start with a full given name ("Yan Zhao")

How groups and contrasts are built

labels: group lead's pipeline on the same Qwen3-32B · we use them unchanged

open answer -> R0 / R1 / R2

greedy, chat template

fuzzy name match: M, H, U

HR2 = (H+U) / (M+H+U)

R2: HR2 <= 0.5 · R0: M = 0 · R1: M > 0

a fixed rule, no human or model judge

multiple choice -> target / control

true list + 3 real teams of other papers

(same field, team size, citation bin, ~year)

4 orders · p(letter) from next token

target: 4/4 and mean p(true) >= 0.5

control: <= 1/4

5 false titles per paper

same field, own year kept

min-cost on |dyear|/5 + |dauthors|

+ |dtitle tokens| + 1.5 |dcitation bin|

no shared surname, Jaccard <= 0.6

79% strict tier (Stage 10, 35,898 of 45,540 pairs, conditions_manifest.json)

statistics

target vs control only within citation bin

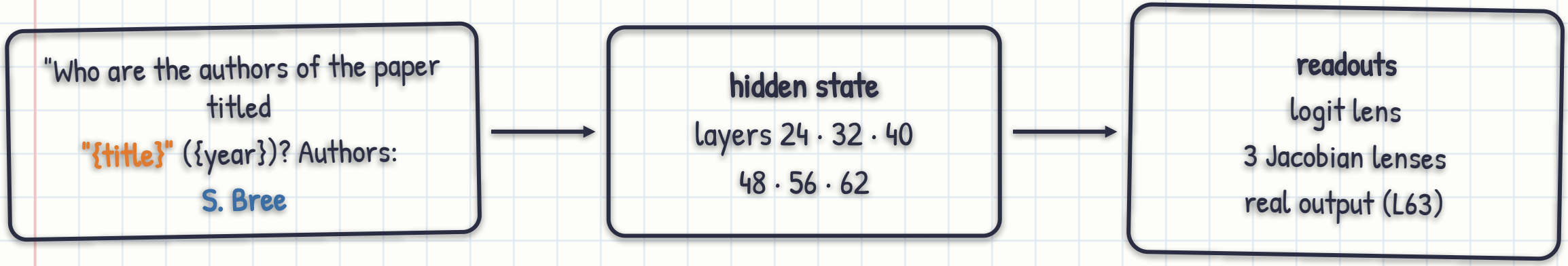
10 bins with >= 30 per group (Stage 10, §6.2)

weight n_t n_c / (n_t + n_c)

95% bootstrap over papers, 2,000

sensitivity: drop bins 0-4

How we read the model



Score = rank of the correct next name token,

as log10(1 / rank): 0 = rank 1, -1 = rank 10

7 title conditions: the true title, 5 matched false titles, no title

title-specific signal

= true title - mean(5 false titles)

Same name, same tokens: what is left is knowing

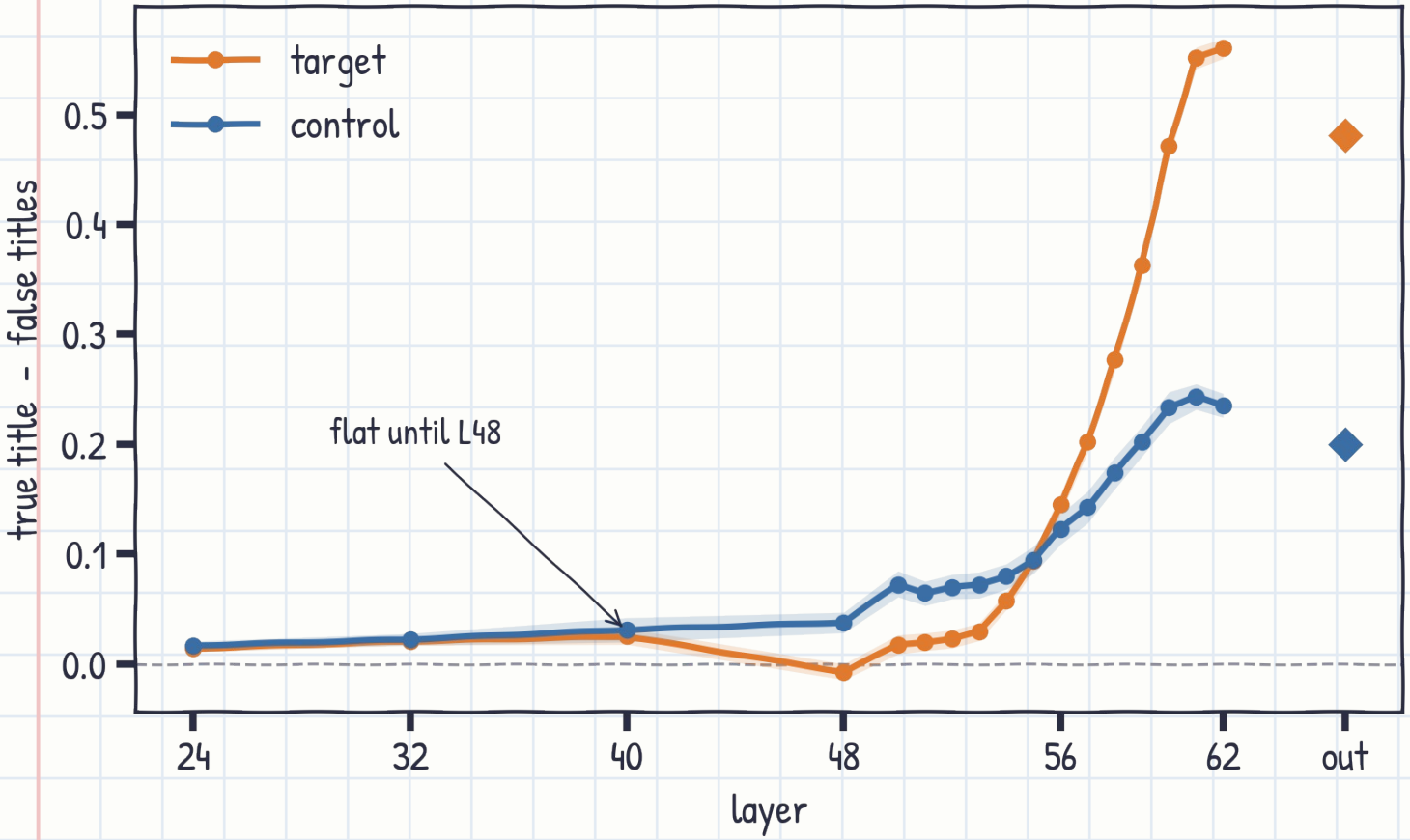
this paper

Lenses (model weights never change; Jacobians to L63 averaged over prompts): public = 557 WikiText-103 prompts (Neuronpedia) · self100 = 100 frozen WikiText-103 prompts · fitA = 100 other WikiText-103 prompts (Stage 9: one skipped prompt replaced -> 100 valid) (n_prompts: 351_stage10/config/lenses.json; Stage 9 final_status.json)

Stage 10

Knowing the paper shows up only at the end

title-specific signal · public Jacobian lens (lines) and model output (diamonds)



L24–L48: ~0 for both groups (Stage 10, all readouts max 0.04, §6.3)

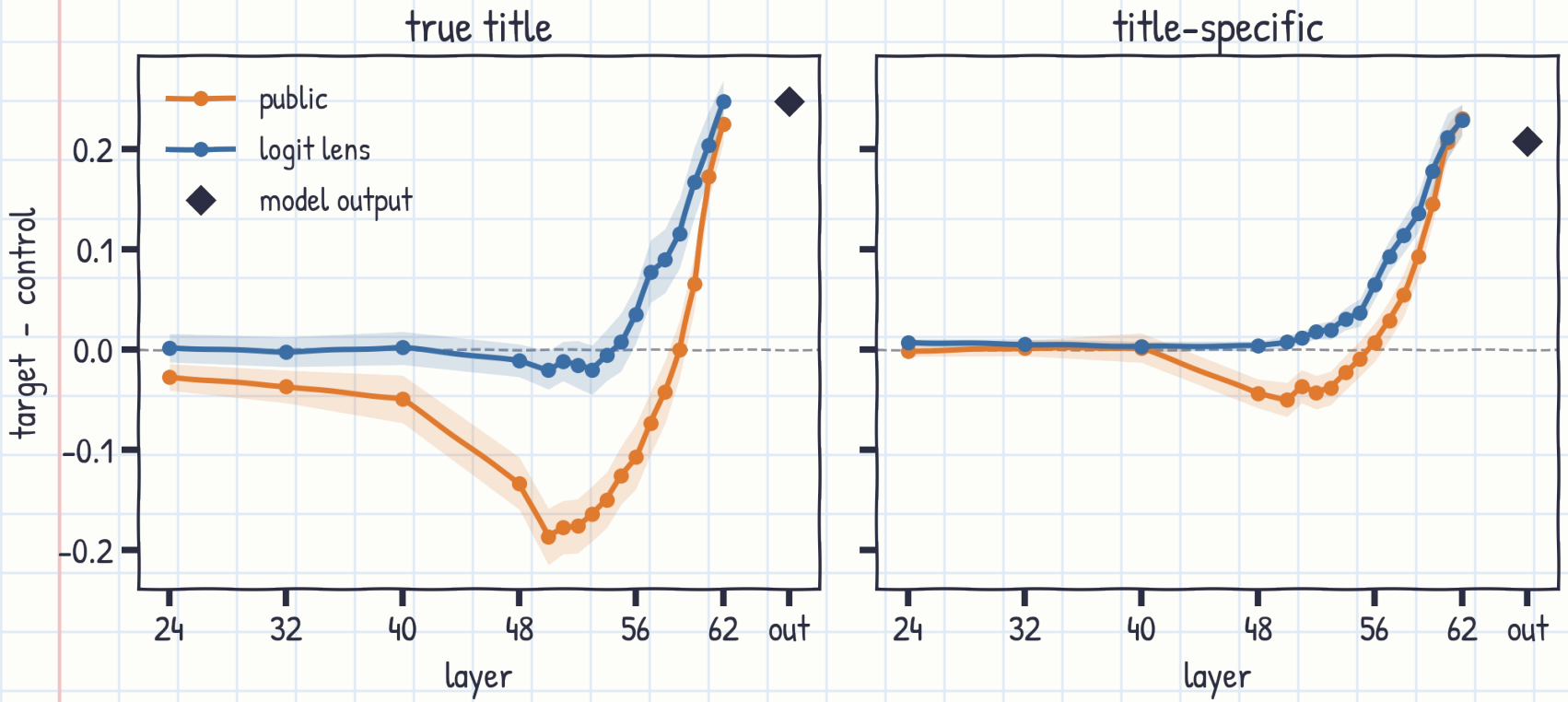
rises at **L56**

L62: **target +0.56** vs **control +0.24**
(Stage 10, public lens L62, 4,621 + 2,220 papers, §6.3 fig.)

Stage 10

Target pulls ahead only in the last layers

target − control, within citation bin, 95% CI



Middle layers: **no head start**

L62 / output:

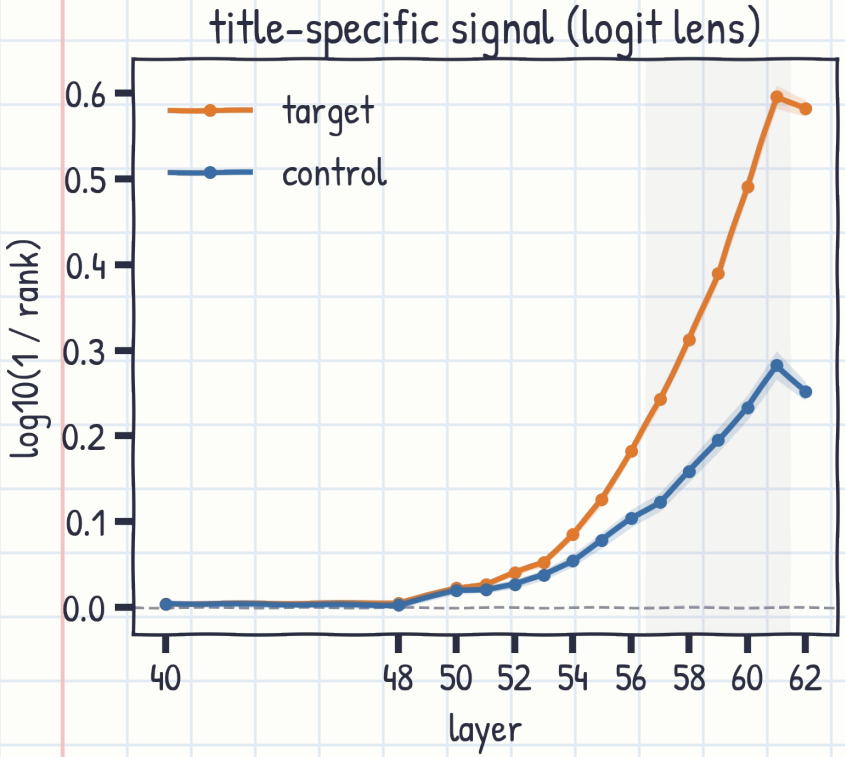
+0.21 to +0.23 (Stage 10, L62 + final L63, 3,734 + 2,182 papers, §6.4)

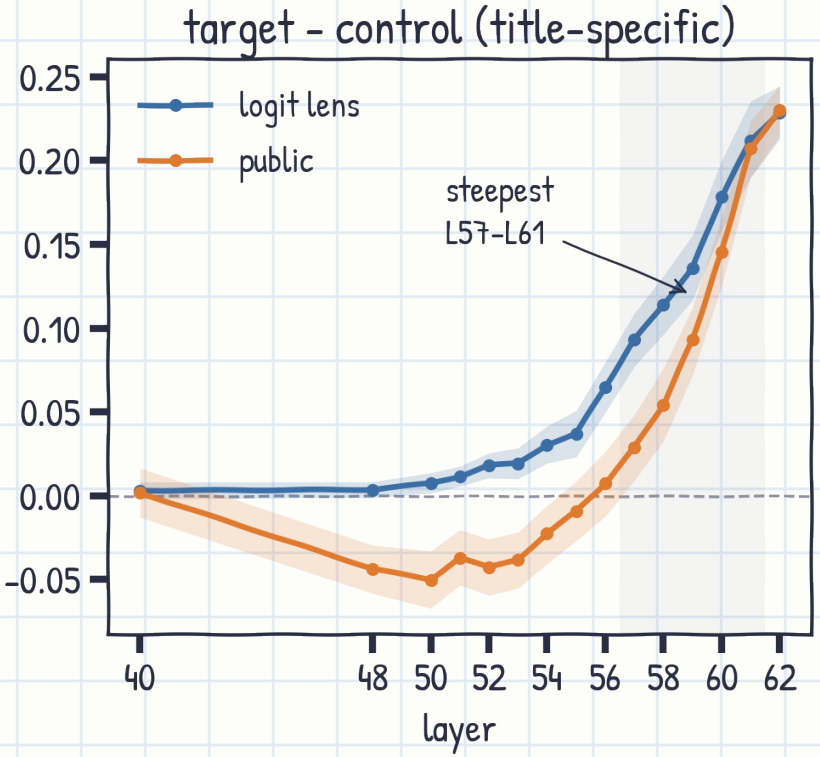
almost all title-specific

R0 only (no author written): output +0.199 vs +0.207 (Stage 10, 3,423 + 2,151 R0 papers, §6.13)

It builds up over the last dozen layers

fine layers L40–L62 · title-specific signal · 95% CI





Logit lens: target − control

+0.007 at L50

+0.23 at L62 (Stage 10 logit lens, 3,734 + 2,182 papers, §6.10)

steepest rise **L57–L61**

Stage 11: intervene, do not just read

593 + 593 papers: min(60, n_t, n_c) per citation bin, seed 11 (Stage 11 sample.parquet, §6.11)

patch uses false_0 · knockout masks all heads · all 9,108 for probe and chain (Stage 10 manifest.json)

patch

copy the true-title hidden state into the false-title run at one layer

knock out

block attention to the title tokens (all heads) from layer l to the last layer

probe

linear map from the last title token to the output state

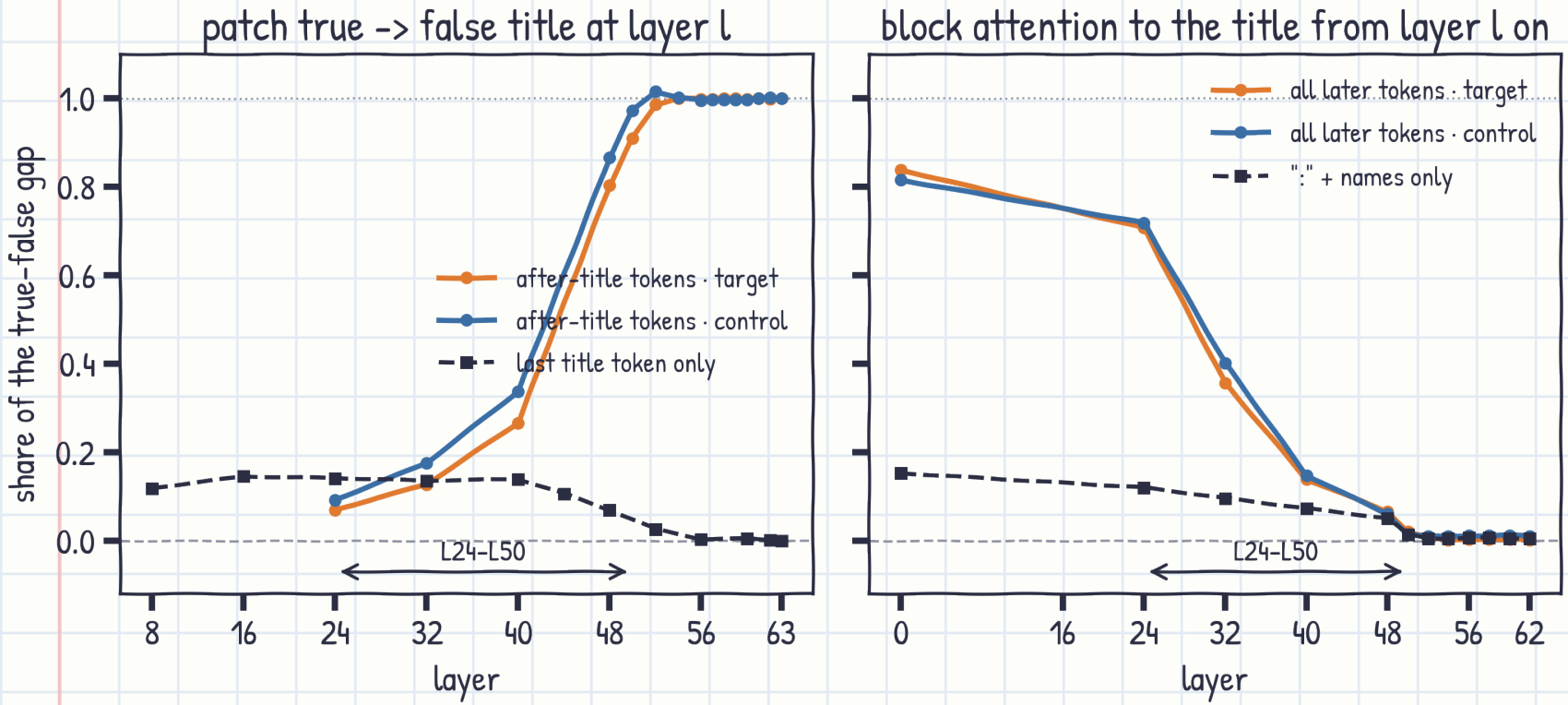
chain

"Authors: A1, A2, ..." instead of each author alone

Score: rank of the correct name token at the model output; "share of the true-false gap" = (intervened − false) / (true − false).

The title is read in L24-L50, early

model output · share of the true-false gap · CIs about +/-0.02 (Stage 11 E1/E2, 593 + 593 papers, §6.11)



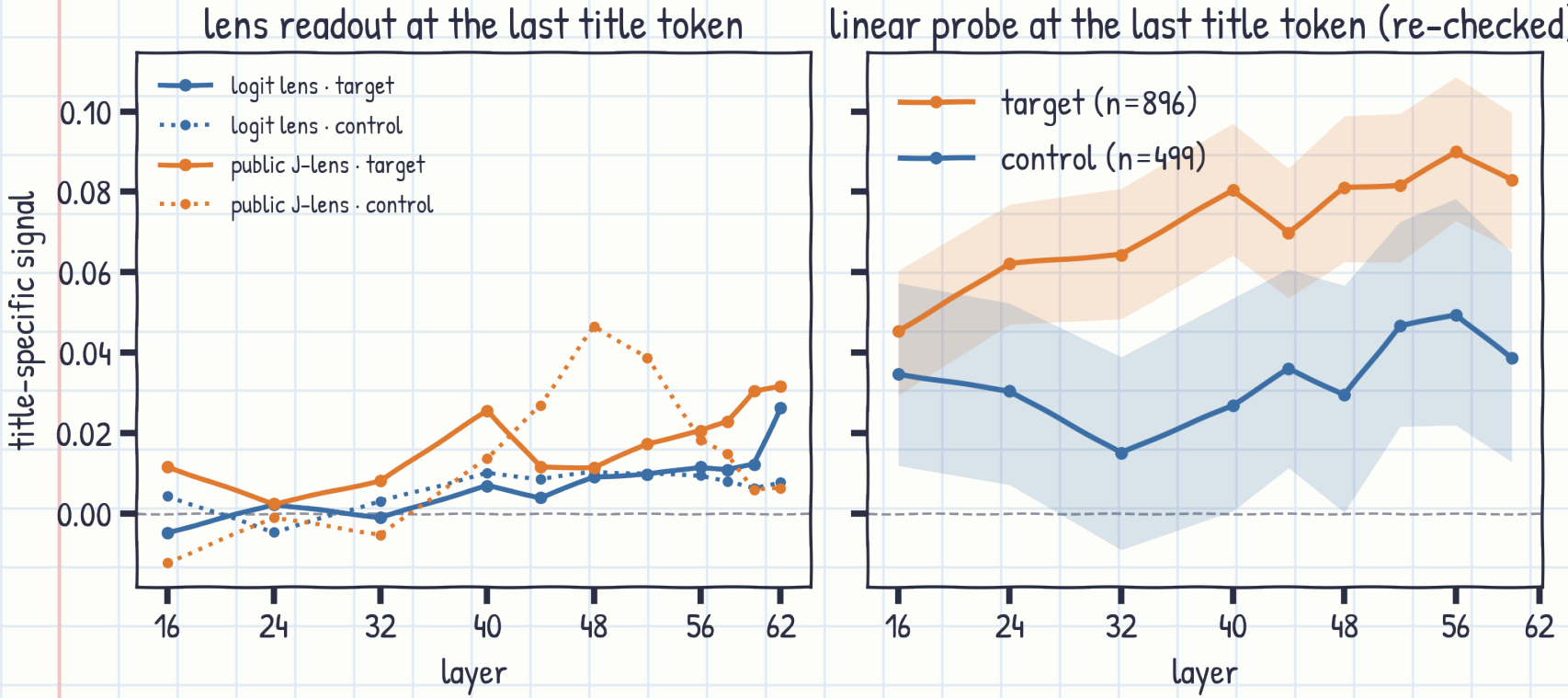
L52+: title no longer needed

the information enters the **name positions at L44-52**, partly via the tokens right after the title

same timing for target and control

Read early: a probe finds it at the title

title-specific signal at the last title token · lenses: all 9,108 papers · probe: re-checked, title reuse removed: 1,863 papers (896 target, 499 control) (Stage 11 E4; Stage 13 E4b)



lenses: **~0.01–0.03** (Stage 11 E4, 4,621 + 2,220, §6.11)

probe: **0.05 -> 0.08** (Stage 13 E4b, 1,863 papers, first name token only)

target > control by **+0.03–0.08** (E4b, 896 vs 499, within bin)

re-checked: team folds, title reuse removed

scores the first name token only (mostly an initial); subtracts false_0 only

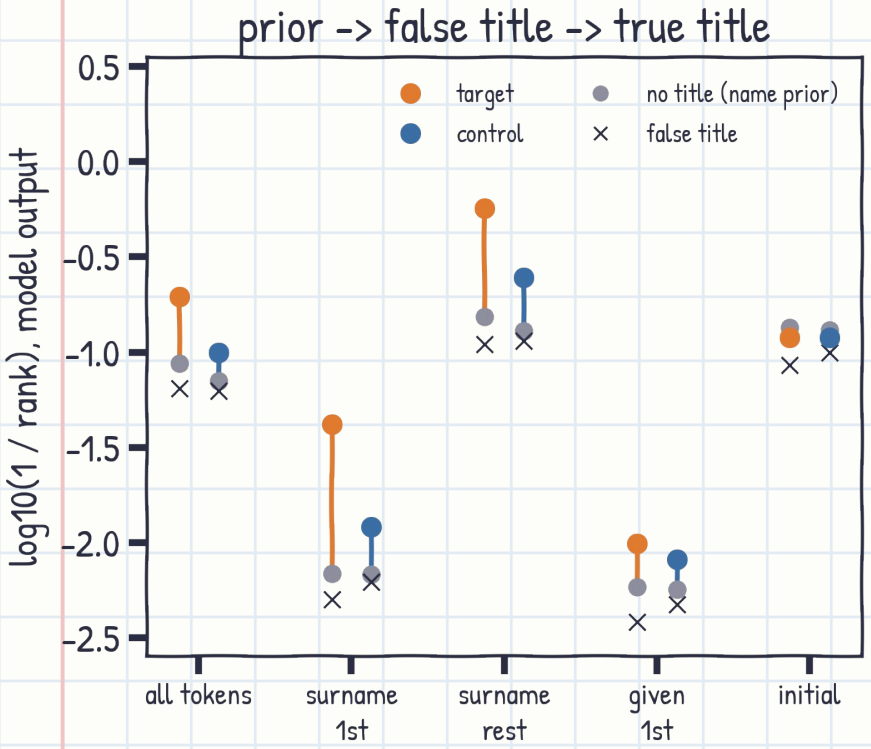
read early (title, from L24), **spoken late** (names, L57–61)

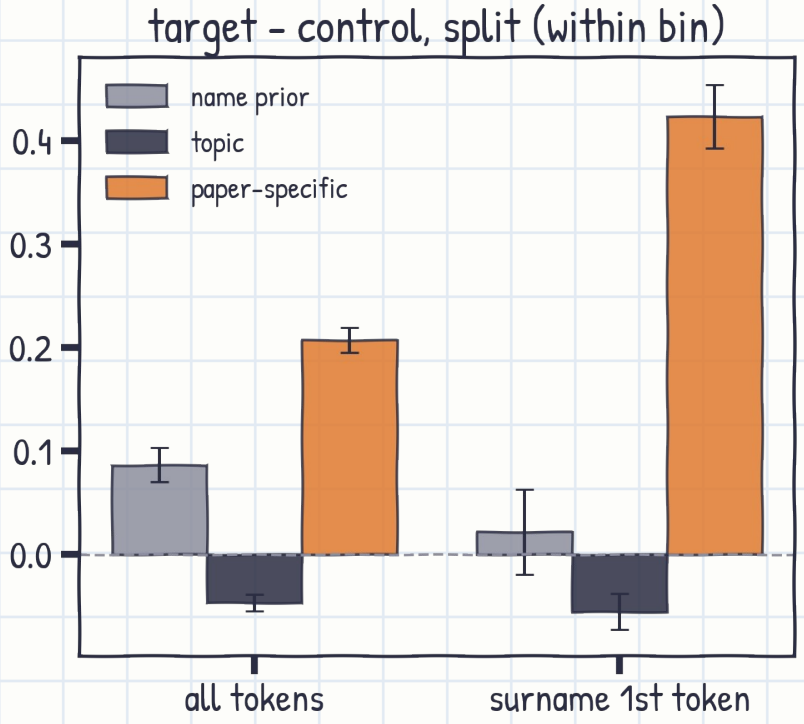
Probe scores only the first name token (mostly an initial); all-token, by-role version (addendum 9, done): surname first token L40 +0.071 [0.058, 0.085] (1,395 papers; e4b_alltok.md).

Stage 11

How much is just the name prior?

model output · no title vs false title vs true title · all 9,108 papers





no title: rank ~12 (Stage 12 A, 9,108 papers, §6.12)

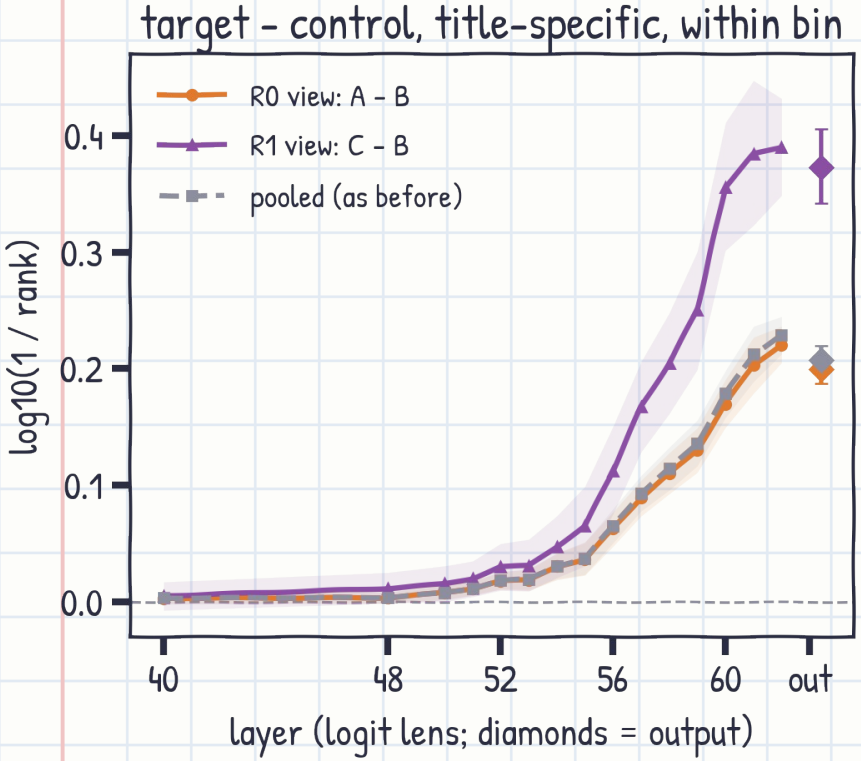
target - control: +0.09 prior, +0.21 paper (Stage 12 A, 4,621 + 2,220, within bin, §6.12)

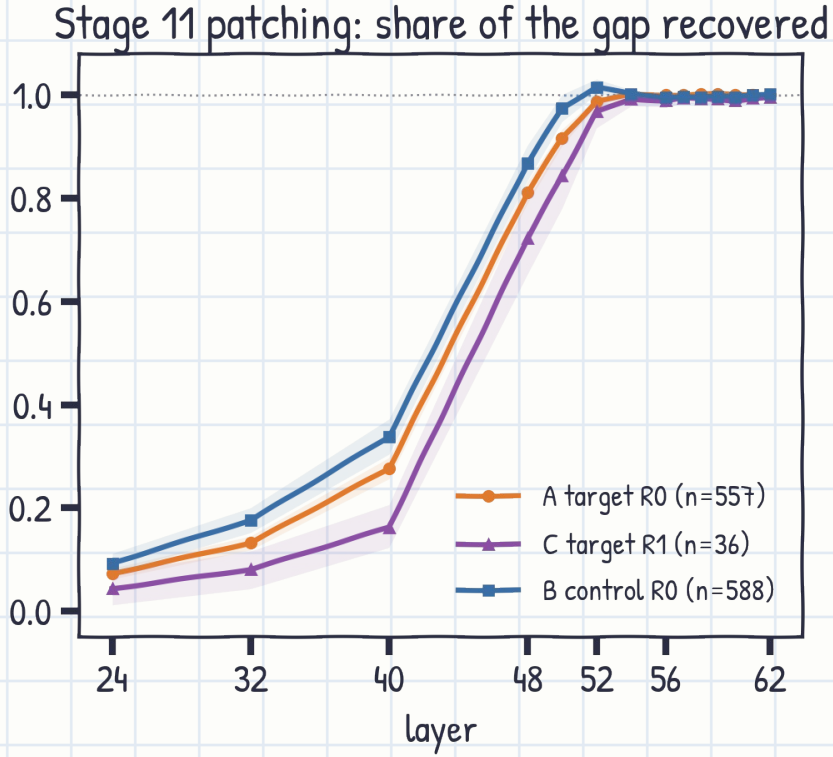
surname 1st: prior ~0, paper +0.42 (same, 1st surname token)

prior vs recognition rho 0.09 (Stage 12 B, 8,894 R0/R1 papers)

Is it just the R1 papers? No.

R0 = no true author written · R1 = a few written · A/B/C/D = target/control x R0/R1





R0 only (A - B):

output +0.199 vs +0.207, same shape (Stage 10, A 3,423 / B 2,151, §6.13)

R1 (C - B): +0.37, also late (C 391 / B 885)

C - A within target: +0.11, from L56 (C 485 / A 3,204)

patching: R1 a bit later, done by L52 (Stage 11 E1, 36 R1 targets)

What we learned: read early, spoken late

Read early

Causally the title is read in L24–L50 and gathered at the name positions by L52, then no longer needed. A probe at the title token reads a little of it from L24 (re-checked).

Spoken late

At the name positions, paper-specific author knowledge becomes lens-readable only from ~L50, fastest over L57–L61.

No lens lead

Read with a lens at the names, target is not ahead of control in middle layers; the +0.21 gap is at the end (Stage 10, final L63, §6.4) (probe at the title: target − control +0.03–0.08, re-checked (Stage 13 E4b, first name token only)).

Real knowledge

Name prior is ~1/3 of the gap overall but ~0 on surnames (Stage 12 A, 0.087 / 0.247, §6.12); the rest is paper-specific and grows with citations.

Lenses agree

Jacobian lenses read more in middle layers, but nothing target-specific.

Name format

Initials carry little; full given names carry as much as surnames and help predict them.

Limits

Done: Stage 12

- Single-position patching (P1); attention vs MLP (P2, hook check passed)
- Name priors (N1, N2); author-prompt lens (F1, F2): middle layers still not readable

Done: Stage 13

- Probes read year, citations, field at the title from ≤L16 (M2); identity not above the L4 text level (M3)
- Replacement is task-independent (X1); paraphrase meaning beyond wording in L28-L40, not separable from paper identity (X3)
- First author decodable at L8, may be text-explainable (X4); 5 false titles confirm the Stage 11 timing (X6, X7)

Limits

- One model (Qwen3-32B)
- Teacher-forced preference, not generation
- Labels from the team lead's package; labelling code (elm) not available
- H2 (paper identity) partly supported but not established; H3 (hidden author knowledge): insufficient evidence

Appendix

checks · readout levels · author chain · probe controls · lenses · citations · name formats

Can we trust the numbers?

fast = slow

One forward pass vs one per prefix: max gap 0.37 logit (bf16 rounding); ranks 1-41 identical (Stage 10, 12 contexts, parity.json)

matches Stage 6

972 tokens: same token ids, median log-prob gap ~1e-5 (Stage 10, crosscheck_stage6.csv)

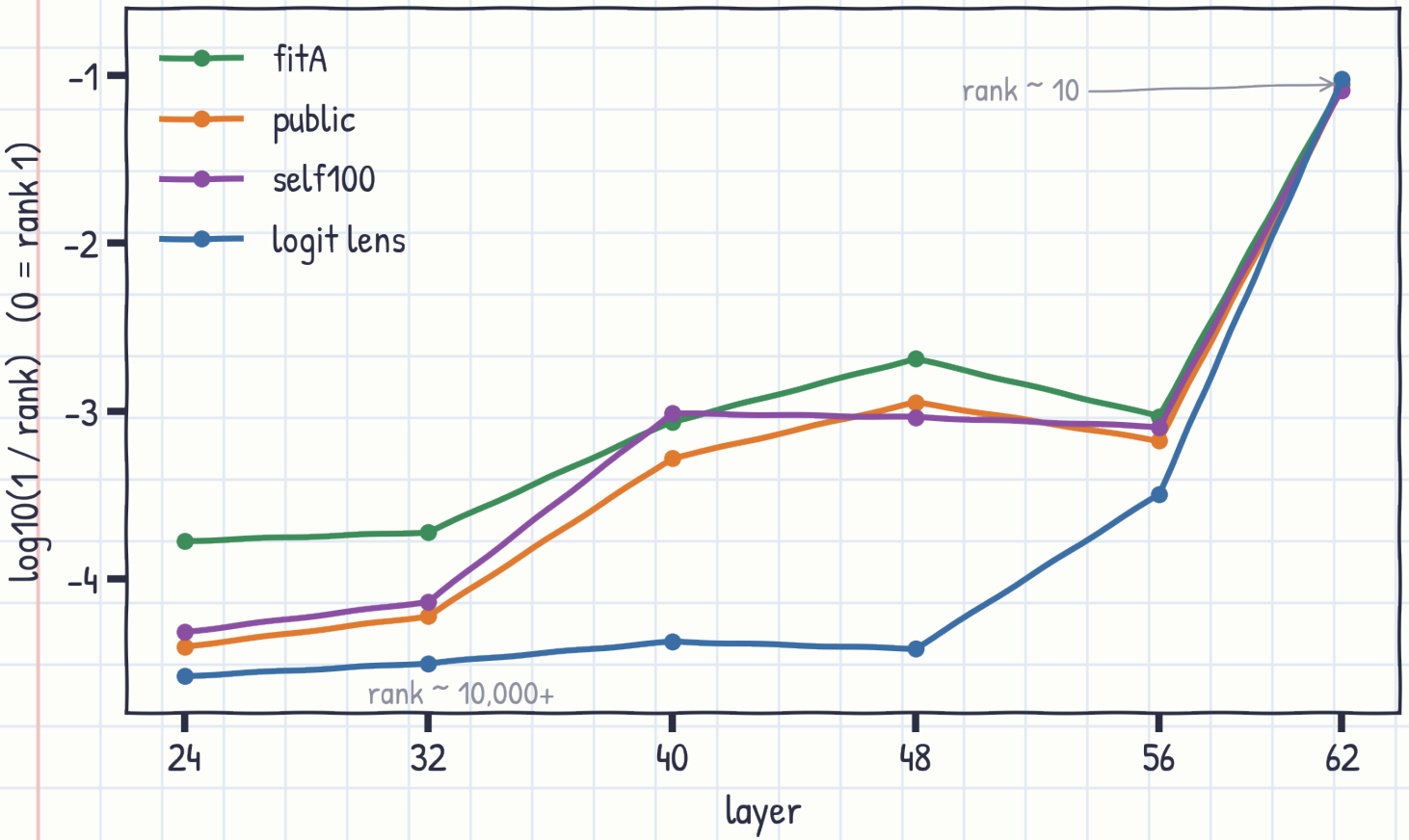
negative control

Punctuation tokens show ~0 title-specific signal at every layer (Stage 10, max 0.014, 9,108 papers, §6.5)

Caveat: the parity tolerance was raised 0.25 -> 0.5 after seeing these bf16 gaps. (parity.json tolerance_amendment)

Names become readable late

all 9,108 papers · true title (Stage 10 cohort, §6.1)



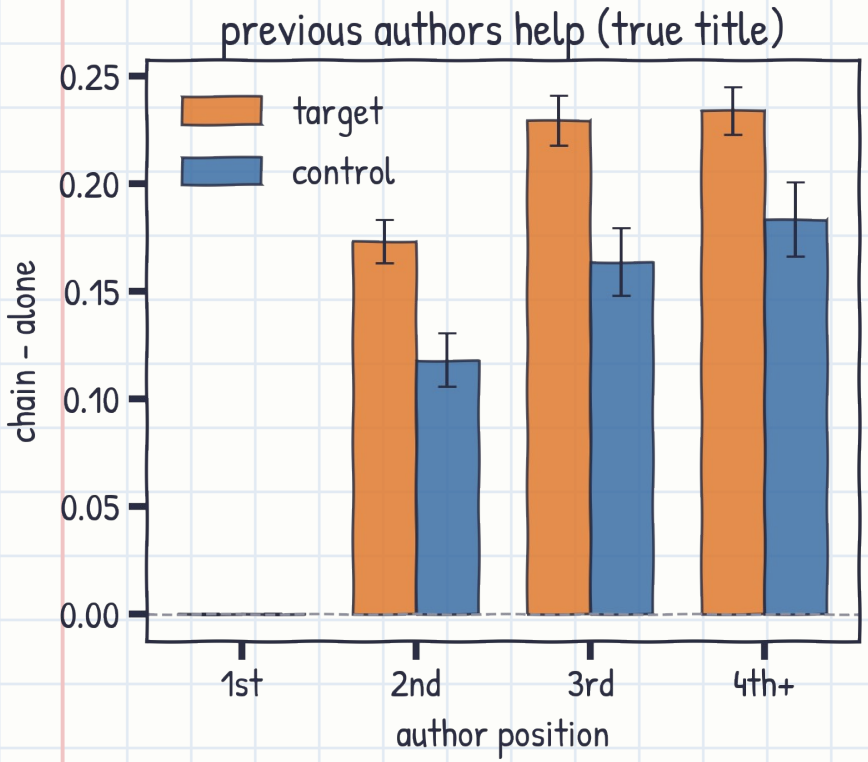
Logit lens: almost nothing until **L48**

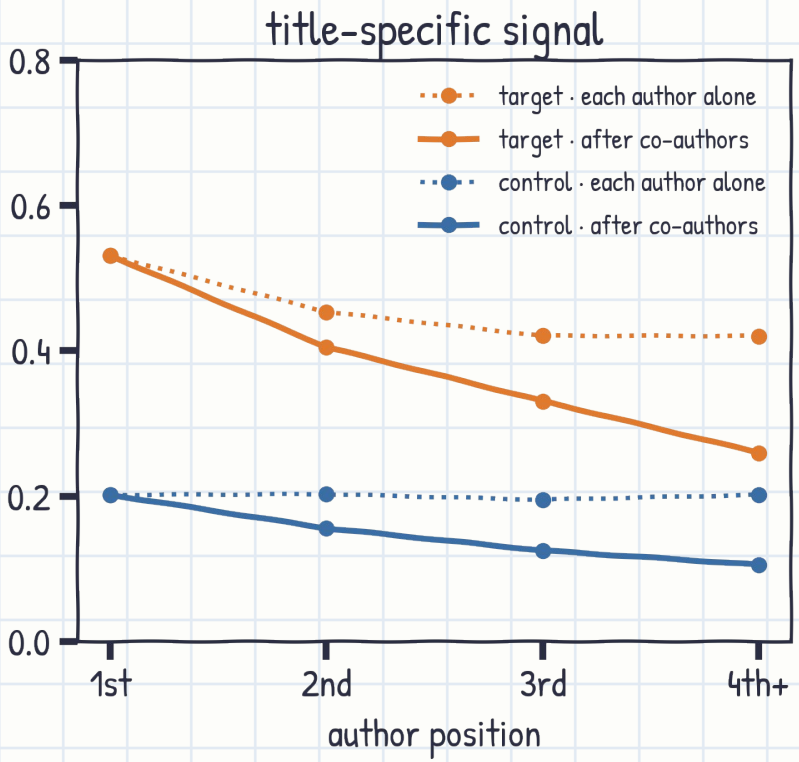
Jacobian lenses read more from **L40**

Everything jumps at **L62** (rank ~10) (Stage 10, logit L62 -0.93, §6.1)

Co-authors help, and partly replace the title

model output · all 9,108 papers · 95% CI (Stage 11 E3, §6.11)





later authors **+0.17 to +0.23** (Stage 11 E3, 4,621 target, §6.11)

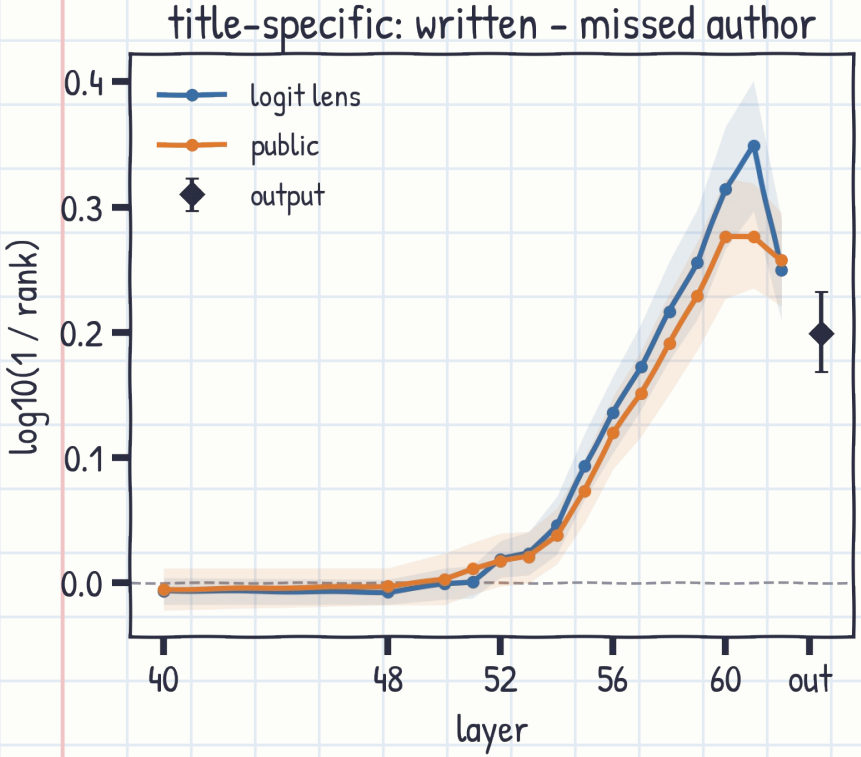
title-specific part **shrinks**

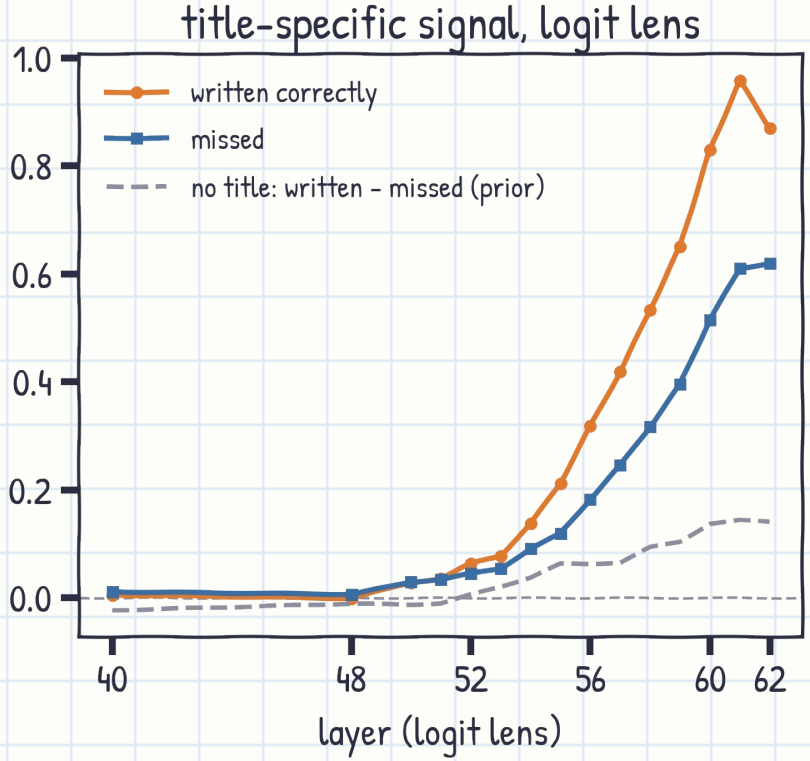
target > control still

Stage 10–11

Authors it wrote vs authors it missed

inside 495 R1 papers · paired within paper · 95% CI (Stage 10, r0_r1_analysis.md §4)





written − missed, output:
+0.20 (Stage 10, final L63, 495 papers)

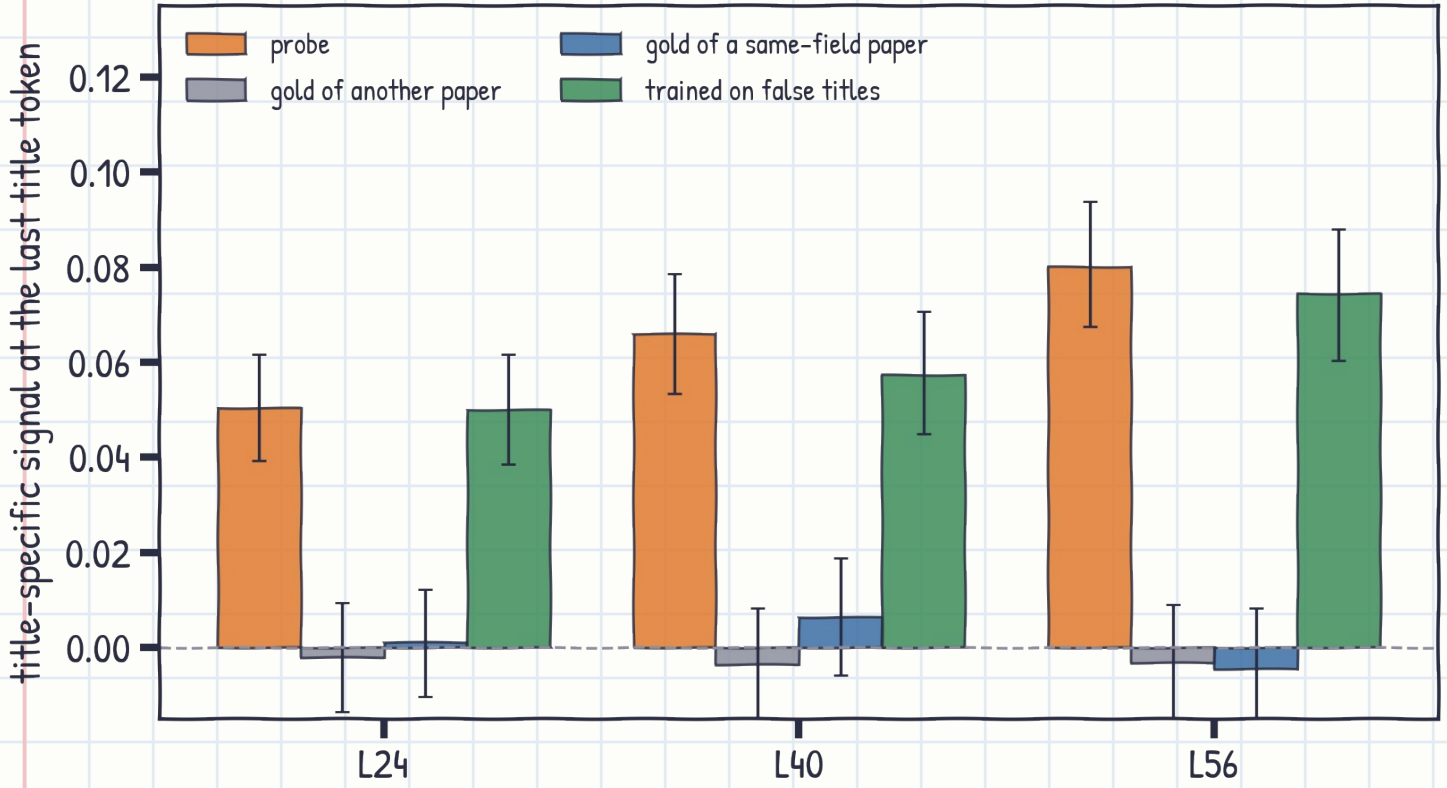
~0 up to L52, grows from **L54**

no title (prior): +0.10 (same, no title)

same without first authors

The probe signal is paper-specific

last title token · 5-fold ridge probe · re-checked: author-team folds, title reuse removed



another paper's authors: **0**

same-field paper: **0** (Stage 12 X1 + Stage 13 re-check, §6.12)

so it is **this paper**, already at L24

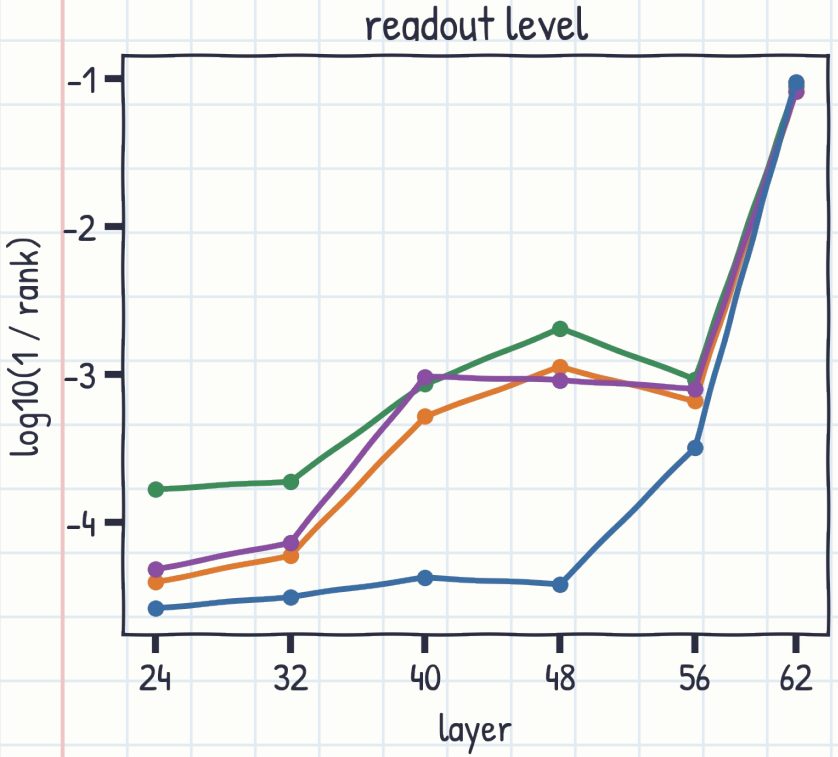
lenses cannot read it

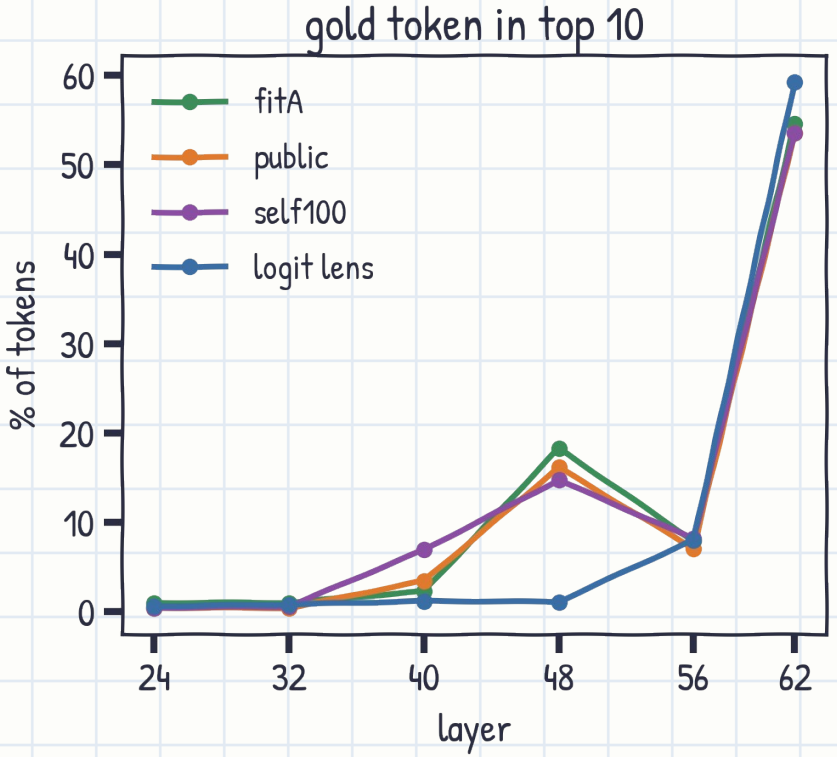
Probe scores only the first name token (mostly an initial); all-token, by-role version (addendum 9, done): surname first token L40 +0.071 [0.058, 0.085] (1,395 papers; e4b_alltok.md).

Stage 10

Which lens reads best?

all papers · true title





Middle layers:

fitA best; at L48 15–18% top-10 vs 1% for logit

(Stage 10 X3, 9,108 papers, §6.9)

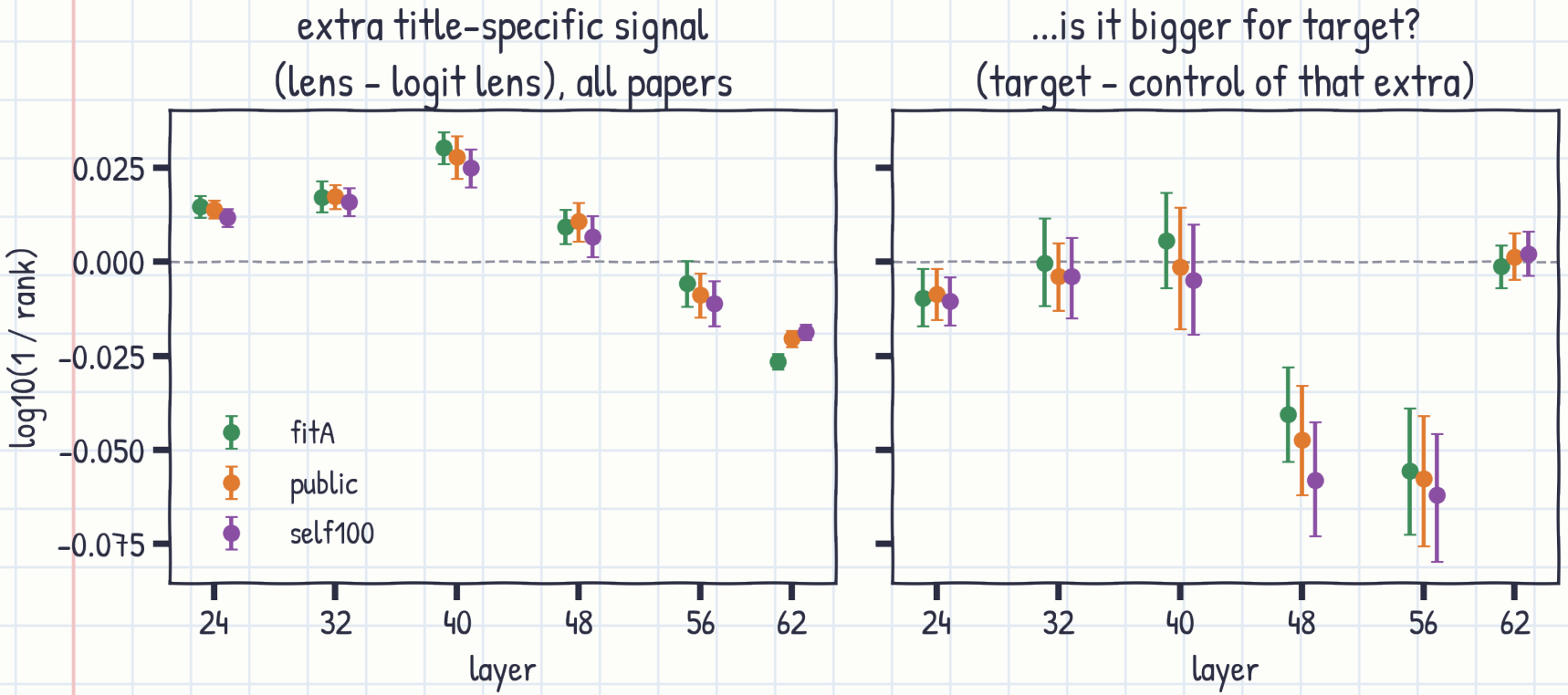
L62: logit lens best (59%)

(same, L62)

Stage 10

Do lenses reveal hidden paper knowledge?

paired differences, 95% CI



Extra signal: **+0.01 to +0.03** (Stage 10 X3, L24–L48, 9,108 papers, §6.9)

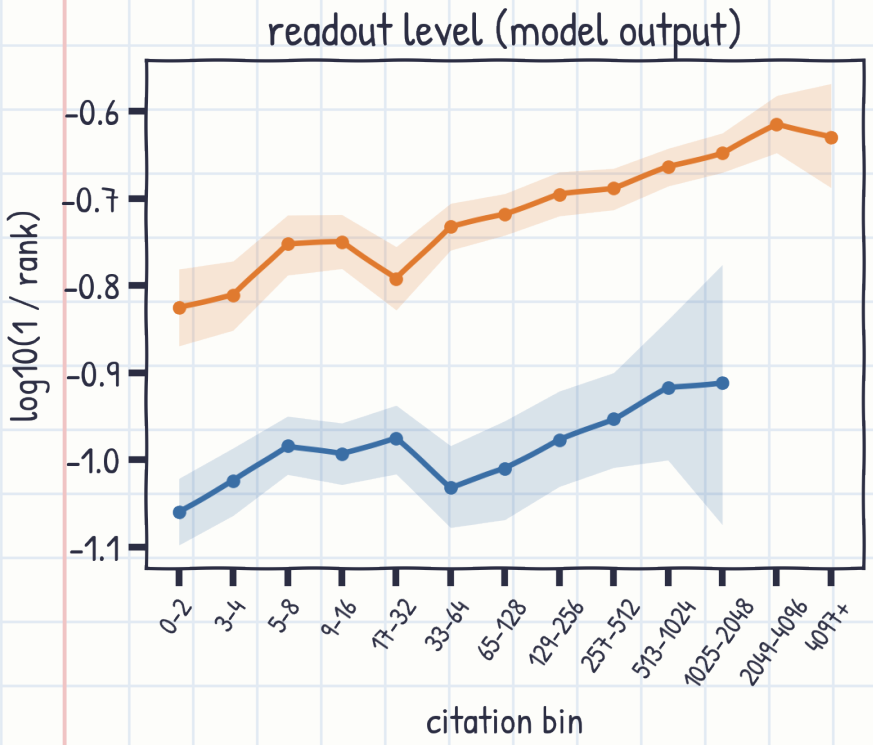
vs ~0.44 at L62 (same, L62)

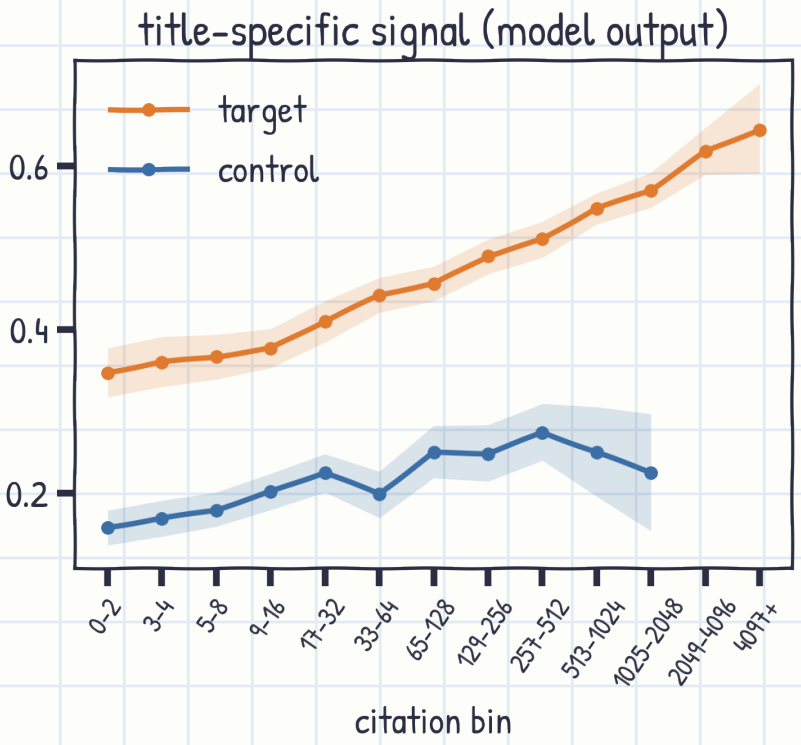
Not bigger for target
(even negative at L48–56)

Stage 10

More citations, more paper knowledge

model output · bins with < 10 papers omitted





Target: +0.35 -> +0.64 (Stage 10, final, 200 / 84 papers, §6.8)

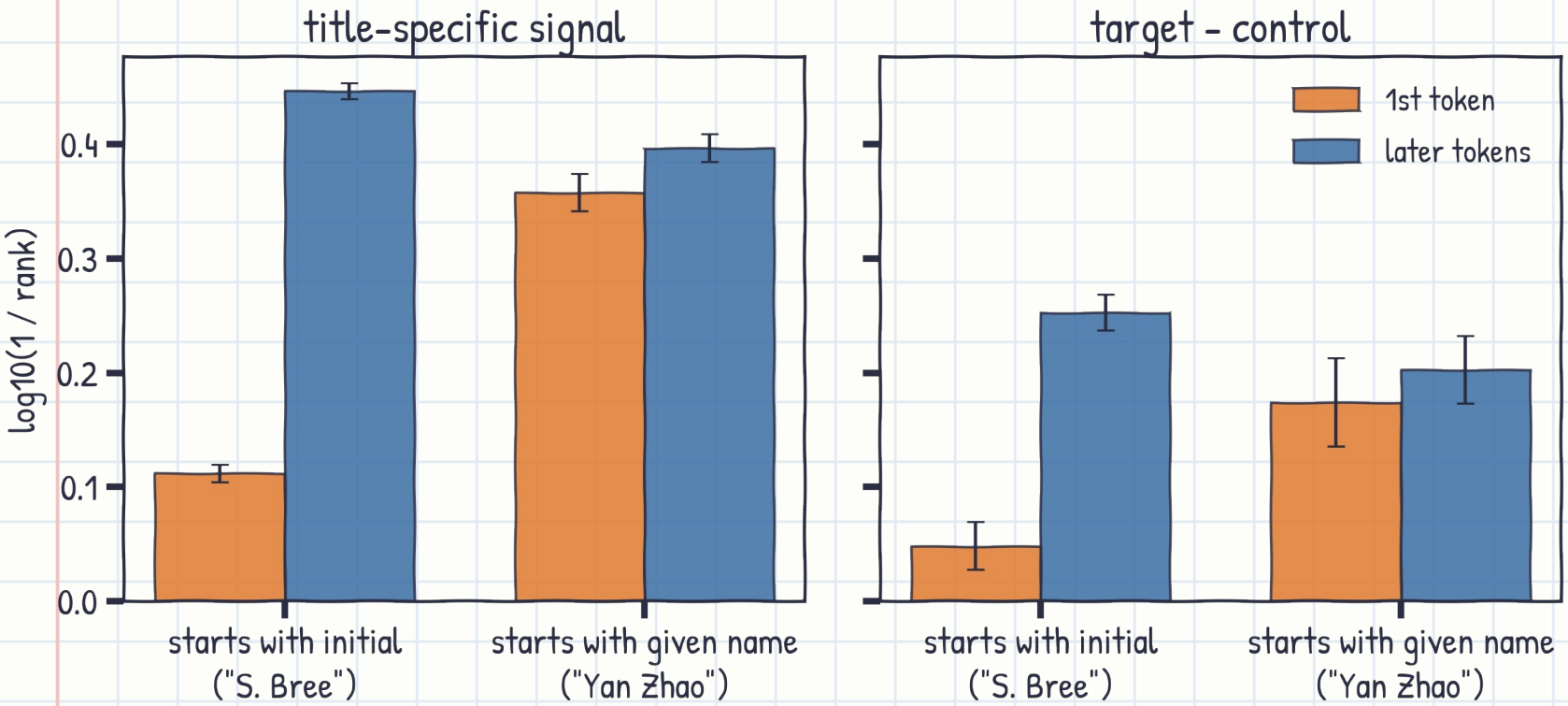
Spearman (title-specific)

0.45 all · 0.33 target · 0.19 control (Stage 10 X2, 9,108 papers, §6.8)

Target > control in every bin

First token or the rest of the name?

model output · split by name format

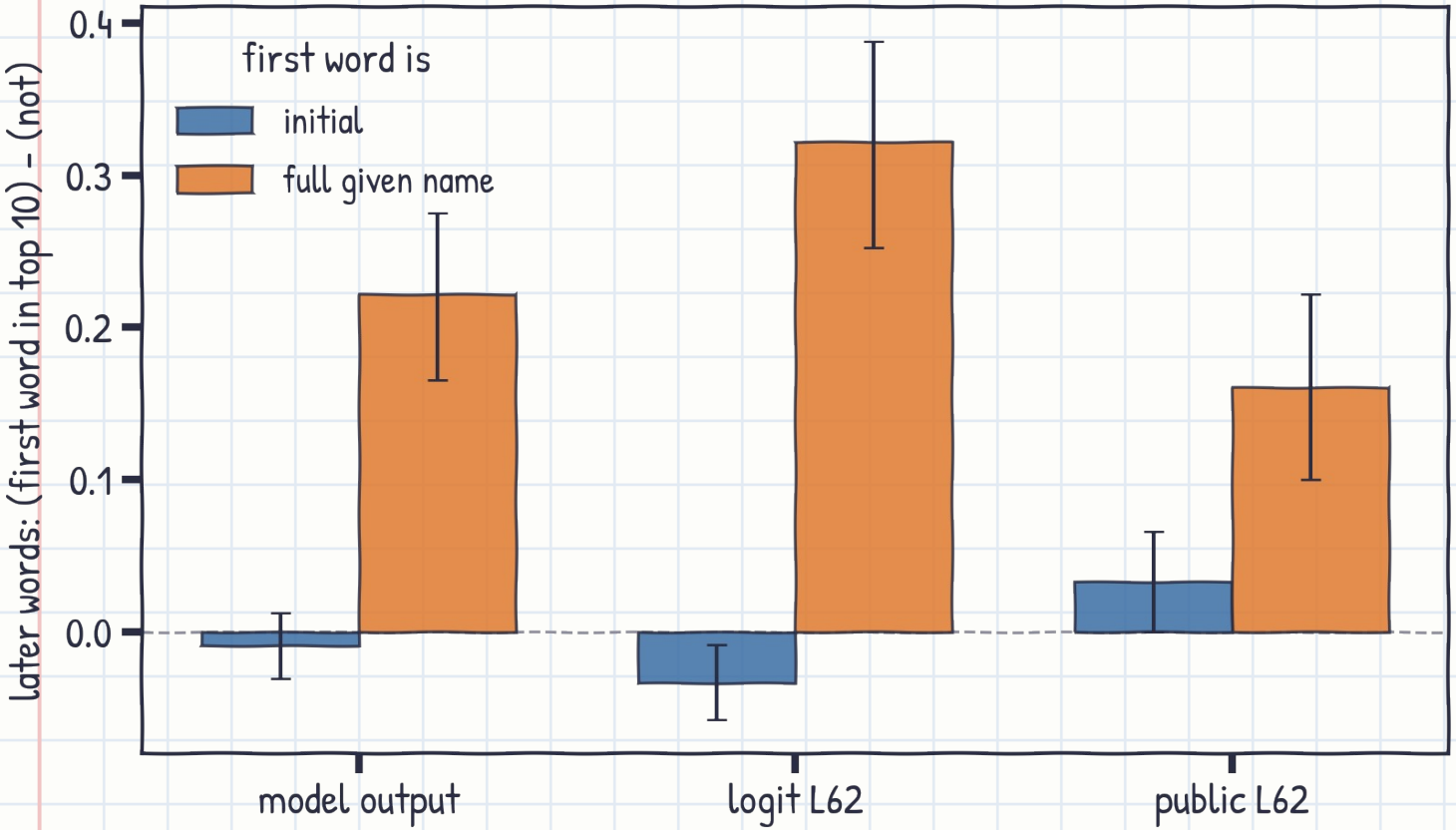


Initial: carries little; signal is on the surname

Full given name: carries about as much as the surname

Knowing the given name helps with the surname

true title · later words when the first word is / is not in the top 10



Given name in top 10

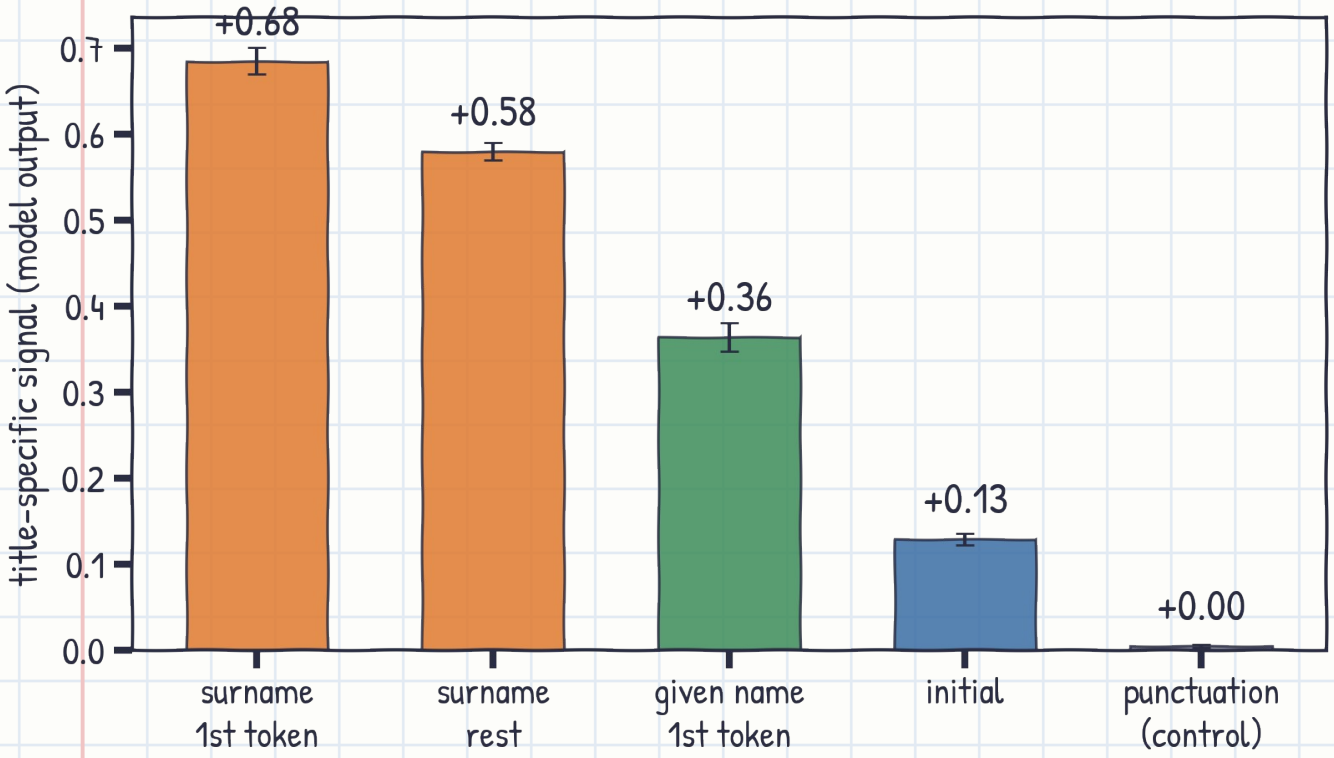
surname +0.22 (output) (Stage 10, 8,876 authors, §6.7)

+0.32 (logit L62) (same)

Initial in top 10: no help

Where the signal sits, and a dose-response

model output





Right: R0/R1 papers, centred within citation bin; Spearman rho = 0.27, $p < 0.001$. (Stage 10, 8,894 papers, §6.6)